

VulnPulse Operations Guide

Automated Vulnerability Assessment Orchestration & Triage Engine

PROJECT DOCUMENTATION v1.0

1. Introduction & Objectives

VulnPulse is a modular Python 3 security orchestration utility engineered to streamline network reconnaissance, service fingerprinting, and HTTP security posture auditing. By wrapping raw tools like Nmap and automating HTTP header inspections, VulnPulse normalizes disparate telemetry into a unified data dictionary and generates structured triage reports.

2. Core Architecture & File Tree

The system utilizes a decoupled, component-based design pattern. The central engine controls execution flow while importing specialized modules and reporting utilities.

```

vulnpulse/ ├── modules/ | ├── __init__.py | ├── recon.py # Handles automated Nmap
execution and XML tree parsing | └── http_audit.py # Handles HTTPS/HTTP fallback and
security header analysis ├── utils/ | └── reporter.py # Aggregates telemetry, calculates
risk scores, and exports JSON reports ├── .gitignore # Excludes cache, temporary XMLs, and
JSON artifacts └── main.py # Core engine and argument parsing controller
```

3. Installation & Setup

Ensure Python 3 and Nmap are installed on your system (tested on Kali Linux environments).

```

# Clone repository or initialize locally git clone
https://github.com/spacevcowboy/vulnpulse.git cd vulnpulse # Install required HTTP
automation libraries pip install requests
```

Operational Note: Always ensure you have explicit authorization before executing network reconnaissance or security assessments against target hosts.

4. Execution & Usage

Run the core orchestrator via the command line by supplying an authorized target domain or IP address using the `-t` or `--target` argument:

```
python3 main.py -t scanme.nmap.org
```

5. Sample Execution & Output Trace

Below is a capture of a live assessment run, showcasing the multi-phase progression from port discovery to automated header auditing and risk scoring:

```
[...] Automated Vulnerability Orchestrator & Triage Engine v1.0 [*]
Starting automated assessment against: scanme.nmap.org --- [ Phase 1: Network & Port Recon ]
--- [*] Executing live Nmap service scan on scanme.nmap.org... [+] Discovered open port:
22/tcp (ssh) [+] Discovered open port: 80/tcp (http) --- [ Phase 2: HTTP & Application Audit
] --- [*] Inspecting HTTP/HTTPS security posture for scanme.nmap.org... [*] Attempting
connection to https://scanme.nmap.org... [!] Connection failed or timed out for
https://scanme.nmap.org [*] Attempting connection to http://scanme.nmap.org... [+] Connected
successfully! Status Code: 200 [+] Security Headers Present: 0/5 - Missing: Strict-
Transport-Security - Missing: Content-Security-Policy - Missing: X-Frame-Options - Missing:
X-Content-Type-Options - Missing: X-XSS-Protection --- [ Phase 3: Generating Executive
Summary ] --- [*] Processing telemetry for scanme.nmap.org...

===== VULNPULSE EXECUTIVE SUMMARY
===== Target Host : scanme.nmap.org Timestamp :
2026-09-23 15:19:52.400268 Risk Rating : Medium Risk Factors : - Multiple security headers
missing (5/5) ----- Open Ports : 2 detected [+]
22/tcp -> SSH [+] 80/tcp -> HTTP ----- Web URL
: http://scanme.nmap.org HTTP Status : 200 Server Tech : Apache/2.4.7 (Ubuntu) Headers
Deficit: 5/5 missing ===== [+] Detailed scan
artifact saved to: report_scanme_nmap_org.json
```